THE UK UNIVERSITY INTEGRATION BEE 2024/2025

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ROUND THREE CHAINLINK ROUND GRADERS' COPY

Sponsored by



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1 Game 1

1. $\displaystyle\int_1^4 \{x\}^{\lfloor x \rfloor}\,\mathrm{d}x$

   Answer: $\dfrac{13}{12}$

2. $\displaystyle\int \frac{\sinh(x)+\cosh(x)}{\sinh(e^x)+\cosh(e^x)}\,\mathrm{d}x$

   Answer: $-e^{-e^x}+C$

3. $\displaystyle\int \frac{1+\ln(x)}{\sqrt{x^{-2x}-1}}\,\mathrm{d}x$

   Answer: $\sin^{-1}(x^x)+C$

   Answer: $-\tan^{-1}\left(\sqrt{x^{-2x}-1}\right)+C$

4. $\displaystyle\int_1^\infty \frac{1}{x\sqrt{x}\sqrt{x-1}}\,\mathrm{d}x$

   Answer: 2

5. $\displaystyle\int_0^1 \frac{\ln(1+x+x^2)}{x}\,\mathrm{d}x$

   Answer: $\dfrac{\pi^2}{9}$

6. $\displaystyle\int_0^1 f(x)\,\mathrm{d}x$, where $f(x)=\begin{cases}1, & x=0.x_1x_2x_3\ldots \text{ where } x_n\neq 9 \text{ for all } n;\\ 0, & \text{else}\end{cases}$

   Answer: 0

7. $\displaystyle\int_{\tan^{-1}\left(\frac{1}{3}\right)}^{\tan^{-1}\left(\frac{1}{2}\right)} \frac{x}{\sin(4x)}\,\mathrm{d}x$

   Answer: $\dfrac{\pi}{32}\left(\ln\left|\csc\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)-\cot\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)\right|-\ln\left|\csc\left(4\tan^{-1}\left(\frac{1}{3}\right)\right)-\cot\left(4\tan^{-1}\left(\frac{1}{3}\right)\right)\right|\right)$

   Answer: $\dfrac{\pi}{32}\ln\left(\dfrac{1-\cos\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)}{1+\cos\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)}\right)$

   Answer: $\dfrac{\pi}{32}\ln\left(\dfrac{1+\cos\left(4\tan^{-1}\left(\frac{1}{3}\right)\right)}{1-\cos\left(4\tan^{-1}\left(\frac{1}{3}\right)\right)}\right)$

   Answer: $\dfrac{\pi}{16}\ln\left(\dfrac{4}{3}\right)$

8. $\displaystyle\lim_{n\to\infty}(n+1)\int_0^1 x^{x^x+n}\ln(1+x+x^x+x^{x^x}+x^{x^{x^x}})\,\mathrm{d}x$

   Answer: $\ln(5)$

2 Game 2

1. $\int_{-1}^{0} 5(x^6+x)^4 \, \mathrm{d}x$

   Answer: $\dfrac{1}{5}$

2. $\int e^{e^{e^x}} e^{e^x} e^{e^x} e^x \, \mathrm{d}x$

   Answer: $e^{e^{e^x}} \left(e^{e^x} - 1\right) + C$

3. $\int_{0}^{\frac{\pi}{2}} \sec(x)\csc(x)\sin(\tan(x)) \, \mathrm{d}x$

   Answer: $\dfrac{\pi}{2}$

4. $\int_{0}^{\infty} \ln(1+e^{-x})\cos(x) \, \mathrm{d}x$

   Answer: $\dfrac{1}{2} - \dfrac{\pi}{2}\operatorname{csch}(\pi)$

5. $\int_{0}^{\frac{\pi}{2}} \dfrac{\sin(x)}{\cos(x)+\tan(x)} \, \mathrm{d}x$

   Answer: $\dfrac{2}{\sqrt{5}} \tanh^{-1}\left(\dfrac{1}{\sqrt{5}}\right)$

   Answer: $\dfrac{1}{\sqrt{5}} \ln\left(\dfrac{\sqrt{5}+1}{\sqrt{5}-1}\right)$

6. $\int_{0}^{\infty} e^{-a^2x^2-\frac{b^2}{x^2}} \, \mathrm{d}x$ for $a, b > 0$

   Answer: $\dfrac{\sqrt{\pi}}{2ae^{2ab}}$

7. $\int_{0}^{\frac{\pi}{2}} \dfrac{\cos^3(x)\sin(x)}{\ln(\sin(x))} \, \mathrm{d}x$

   $-\ln(2)$

8. $\lim_{n\to\infty} \int_{0}^{1} 2025\sin(\{nx^{2025}\})\cos(\{nx^{2025}\}) \, \mathrm{d}x$

   Answer: $\dfrac{2025}{2}\sin^2(1)$

3 Game 3

1. $\int_0^\infty e^{-x^{\frac{1}{2025}}}\,\mathrm{d}x$

   Answer: 2025!

2. $\int_{-1}^\infty e^{-x^2}(5x^4-2x^6)\,\mathrm{d}x$

   Answer: $\dfrac{1}{e}$

3. $\int (x+2x^2+3x^3+\cdots+2025x^{2025})(1+x+x^2+\cdots+x^{2024})\,\mathrm{d}x$

   Answer: $\dfrac{1}{2}(x+x^2+x^3+\cdots+x^{2025})^2+C$

   Answer: $\dfrac{1}{2}\left(\dfrac{x-x^{2026}}{1-x}\right)^2+C$

4. $\int_0^{\frac{\pi}{4}} \dfrac{\sin(2x)}{\sin(x)+\cos(x)+1}\,\mathrm{d}x$

   Answer: $1-\dfrac{\pi}{4}$

5. $\int_0^{\frac{\pi}{2}} \ln(\sin(x))\tan(x)\,\mathrm{d}x$

   Answer: $-\dfrac{\pi^2}{24}$

6. $\int_0^\infty \dfrac{x^\varphi \tan^{-1}(x)}{1+x^{2\varphi^2}}\,\mathrm{d}x$, where $\varphi=\dfrac{1+\sqrt{5}}{2}$ is the golden ratio

   Answer: $\dfrac{\pi^2}{8\varphi^2}$

   Answer: $\dfrac{\pi^2}{2\left(1+\sqrt{5}\right)^2}$

   Answer: $\dfrac{\pi^2}{2(6+2\sqrt{5})}$

7. $\int_0^2 (3x^2-5x+3)\cos(x^3-3x^2+4x-2)\,\mathrm{d}x$

   Answer: $2\sin(2)$

8. $\int_0^{\frac{\pi}{2}} \dfrac{1-\ln(\cos(x))}{x^2+(1-\ln(\cos(x)))^2}\cos(2x)\,\mathrm{d}x$

   Answer: $\dfrac{\pi}{4(1+\ln(2))^2}$

4 Game 4

1. $\displaystyle\int \frac{\sin^2(x)}{\cos^4(x)}\,\mathrm{d}x$

   Answer: $\dfrac{\tan^3(x)}{3}+C$

2. $\displaystyle\int_0^1 \frac{1+\sqrt{1-x^2}}{\sqrt{1+x}+\sqrt{1-x}}\,\mathrm{d}x$

   Answer: $\dfrac{2\sqrt{2}}{3}$

3. $\displaystyle\int_1^\infty \frac{(-1)^{\lceil x\rceil}}{\sqrt{\lfloor x\rfloor\,\lceil x\rceil}\left(\sqrt{\lceil x\rceil}-\sqrt{\lfloor x\rfloor}\right)}\,\mathrm{d}x$

   Answer: 1

4. $\displaystyle\int \sqrt{\frac{\ln(x+\sqrt{1+x^2})}{1+x^2}}\,\mathrm{d}x$

   Answer: $\dfrac{2}{3}\left(\sinh^{-1}(x)\right)^{\frac{3}{2}}$

   Answer: $\dfrac{2}{3}\left(\ln\left(x+\sqrt{1+x^2}\right)\right)^{\frac{3}{2}}$

5. $\displaystyle\int_0^{\frac{\pi}{4}} \frac{\cos(2x)}{-\ln(\tan(x))\cos^4(x)}\,\mathrm{d}x$

   Answer: $\ln(3)$

6. $\displaystyle\int \frac{1}{1+x^2}\tan^{-1}\left(\frac{4x-4x^3}{x^4-6x^2+1}\right)\,\mathrm{d}x$

   Answer: $\dfrac{1}{8}\left(\tan^{-1}\left(\dfrac{4x-4x^3}{x^4-6x^2+1}\right)\right)^2+C$

7. $\displaystyle\lim_{n\to\infty}\int_0^{\frac{1}{2}} f_n(x)\,\mathrm{d}x$, where $f_n(x)=\begin{cases}2x-2x^2, & n=1;\\ (\underbrace{f_1\circ\cdots\circ f_1}_{n\ f_1\text{'s}})(x), & n\geq 2\end{cases}$

   Answer: $\dfrac{1}{4}$

8. $\displaystyle\int_0^\infty \frac{x\cosh(x)}{\sinh(\pi x)}\,\mathrm{d}x$

   Answer: $\dfrac{1}{4}\sec^2\left(\dfrac{1}{2}\right)$

5 Game 5

1. $\displaystyle\int_0^\infty \frac{\tan^{-1}(x)}{x^2+1}\,\mathrm{d}x$

   Answer: $\dfrac{\pi^2}{8}$

2. $\displaystyle\int \frac{\sec^2(x)\csc^2(x)}{\tan^2(x)+\cot^2(x)+2}\,\mathrm{d}x$

   Answer: $x+C$

3. $\displaystyle\int_0^1 \frac{1}{1+\sqrt{\frac{1-x+\sqrt{x}}{x+\sqrt{1-x}}}}\,\mathrm{d}x$

   Answer: $\dfrac{1}{2}$

4. $\displaystyle\int_0^1 \frac{1}{x}-\left\lfloor\frac{1}{x}\right\rfloor-\frac{1}{x^2}\left\lfloor\frac{1}{x}\right\rfloor^{-2}\,\mathrm{d}x$

   Answer: $1-\gamma-\dfrac{\pi^2}{6}$

5. $\displaystyle\int_0^1 \sqrt{1+\sqrt[3]{x-1}}\,\mathrm{d}x$

   Answer: $\dfrac{16}{35}$

6. $\displaystyle\int_{-\infty}^\infty \frac{1}{(x^4+1)^2}\,\mathrm{d}x$

   Answer: $\dfrac{3\sqrt{2}}{8}\pi$

7. $\displaystyle\int_0^{2025} \min\left(\sqrt{\{x\}-\{x\}^2},\left|\frac{\sqrt{3}\arcsin\left(\cos\left(\pi x\right)\right)}{\pi}\right|\right)\,\mathrm{d}x$

   Answer: $\dfrac{2025\pi}{12}$

8. $\displaystyle\int \frac{\cos(x)+1-x^2}{(x\sin(x)+1)\sqrt{1-x^2}}\,\mathrm{d}x$

   Answer: $\sin^{-1}\left(\dfrac{x+\sin(x)}{x\sin(x)+1}\right)+C$